MIL-W-17265/1A(SHIPS)
18 January 1968
SUPERSEDING
MIL-W-17265/1(SHIPS)
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(See 6.2)

MILITARY SPECIFICATION

WINCH, DRUM, POWER OPERATED (HIGHLINE WINCH TYPE 19.5-HL-1D, SPANWIRE

WINCH TYPES 19.5-SW-1D, 15.5-SW-1D, 15.5-SW-2D, 8.0-SW-1D, and 8.0-SW-2D)

1. SCOPE

1.1 Scope. This specification covers the detail requirements for highline and spanwire winches.

1.2 Classification. *Winches shall be classified in accordance with the following types as specified* (see 6.1).

Type 19.5-HL-1D - Single drum highline winches rated 19,500 pounds at 240 feet per minute.
Type 19.5-SW-D - Single drum spanwire winches for double hose fueling stations rated 19,500 pounds at 240 feet per minute.
Type 15.5-SW-1D - Single drum spanwire winches for double hose fueling stations rated 15,500 pounds at 240 feet per minute.
Type 15.5-SW-2D - Double drum spanwire winches for double hose fueling stations rated 15,500 pounds at 240 feet per minute.
Type 8.0-SW-1D - *Single drum spanwire winches for* single hose fueling stations rated 8,000 pounds at 240 feet per minute.
Type 8.0-SW-2D - Double drum spanwire winches for single hose fueling stations rated 8,000 pounds at 240 feet per minute.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
RR-W-410 - Wire Rope and Strand.

MILITARY
MIL-W-17265 - Winch, Drum, Power Operated (Electric-Hydraulic, Variable Speed)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. Winches, drum, power operated, highline and spanwire of the types specified in 1.2 shall be in accordance with MIL-W-17265, except as specified herein.

3.2 Drums. Drums for the various winch types shall be designed for the following:

(a) Type 19.5-HL-1D and 19.5-SW-1D Winches -
(1) Stowage capacity for 800 feet of 1 inch wire rope. The breaking strength of the wire rope is 85,600 pounds.
(2) Barrel diameter of 30 inches plus or minus 1/2 inch.
(3) Distance between flanges shall be 15 inches plus or minus 1/2 inch.

FSC 3950

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(b) Type 15.5-SW-1D and 15.5-SW-2D Winches -
(1) Stowage capacity for 800 feet of 7/8 inch wire rope. The breaking strength of the wire rope is 65,800 pounds.
(2) Barrel diameter of 30 inches plus or minus 1/2 inch.
(3) Distance between flanges shall be 15 inches plus or minus 1/2 inch.
(c) Type 8.0-SW-1D and 8.0-SW-2D Winches -
(1) Stowage capacity for 800 feet of 3/4 inch wire rope. The breaking strength of the wire rope is 48,600 pounds.
(2) Barrel diameter of 24 inches plus or minus 1/2 inch.
(3) Distance between flanges shall be 15 inches plus or minus 1/2 inch.

3.2.1 Wire rope construction. Wire rope specified for the winch shall be Type 1, Class 3, Construction 6, independent wire core (IWRC), improved plow steel, uncoated, preformed, right regular lay in accordance with RR-W-410.

3.3 Controlled spooling.

3.3.1 The drum on type 19.5-HL-1D, 19.5-SW-1D, 15.5-SW-1D and 8.0-SW-1D winches as well as the drum used for the spanwire on type 15.5-SW-2D and 8.0-SW-2D winches shall be provided with controlled spooling. The controlled spooling shall insure that the wire spools on the drum in an evenly spaced, interrupted spiral pattern from flange to flange at all times when the equipment is in operation as well as during start-up after periods of disuse without making adjustments. The first and each succeeding layer of wire on the drum shall form a spooling pattern for the next layer that will exactly reproduce this spooling pattern on all layers. The spooling pattern of wire on the drum shall eliminate overriding (cross-over) turns and undercutting of the wire rope.

3.3.2 This type of controlled spooling shall exist during all periods of winch operation when the winch is used in the tensioned highline or tensioned spanwire system as well as when the winch is used for housefall or burtoning between ships. The requirements of 3.3.2.1 and 3.3.2.2 shall also be complied with.

3.3.2.1 The controlled spooling system shall operate at zero line pull and maximum winch speeds in both haul-in and payout directions.

3.3.2.2 The controlled spooling system shall operate at maximum winch deceleration and maximum acceleration between full speed haul-in and full speed payout for all winch loads from zero to rated load.

3.3.3 Wire rope life requirements. The winch controlled spooling system shall provide at least 24,000 cycles (see 3.3.3.1) of operation without failure (see 3.3.3.3) of the wire rope. Twenty percent of the 24,000 cycles shall be conducted at zero line pull under maximum speed change between full speed haul-in and full speed payout. Eighty percent of the 24,000 cycles shall be conducted at winch rated load and rated speed at the specified cycle rate (see 3.3.3.2).

3.3.3.1 Cycle. A cycle is defined as passage of a section of wire rope from the winch drum through the winch controlled spooling system and back onto the winch drum.

3.3.3.2 Cycle rate. Cycle rate when operating at winch rated speed and rated load is defined as one cycle every fifteen seconds.

3.3.3.3 Wire rope failure. Wire rope failure is defined as seven or more broken wires in one strand in one rope lay.

3.3.4 Controls. The winch controlled spooling system shall be provided with controls as necessary to permit initial spooling of wire rope on the winch and through the shipboard underway replenishment system. If controls for the shipboard underway replenishment operation including passing and returning the rig are required, they shall be located in the underway control station.

3.4 Gypsy heads. Winch types specified in 1.2 shall be provided with one gypsy head.

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3.5 Rated line pull and line speed. Rated line pull and line speed for the various winch types shall be as follows:

(a) Type 19.5-HL-1D and type 19.5-SW-1D Winches.
  (1) Rated line pull - 19,500 pounds on the average layer of the drum.
  (2) Rated line speed - 240 feet per minute.
(b) Type 15.5-SW-1D and 15.5-SW-2D Winches -
  (1) Rated line pull - 15,500 pounds on the average layer of the drum.
  (2) Rated line speed - 240 feet per minute.
(c) Type 8.0-SW-1D and 8.0-SW-2D Winches -
  (1) Rated line pull - 8,000 pounds on the average layer of the drum.
  (2) Rated line speed - 240 feet per minute.

3.6 Duty. Type 19.5-HL-1D and 19.5-SW-1D winches shall be designed to permit continuous operation for a period of twelve hours for the normal operating cycle specified below. Type 15.5-SW-1D, 15.5-SW-2D, 8.0-SW-1D and 8.0-SW-2D winches shall be designed to permit continuous operation for a period of six hours for the normal operating cycle specified below. A normal operating cycle for all of the above winch types will consist of 9 seconds at rated speed rated load; 18 seconds at idle; 15 seconds at half speed and rated load; and 18 seconds at idle.

3.6.1 Name plate ampere rating of the electric motor shall not be exceeded by more than twenty five percent when the winch is being operated at rated load and rated speed (constant velocity), temperature rise shall not exceed name plate permissible temperature rise when the winch is operated at the duty cycle and period specified in 3.6.

3.6.2 Thermal overload. Thermal overload shall be indicated by energizing a master panel indicating light through a thermal heating device (see thermal requirements of MIL-W-17265) in either the motor or the controller. The indicator light shall be in accordance with the illumination requirements of MIL-W-17265. This device shall not de-energize the motor.

3.7 Light line. Light line requirements specified in MIL-W-17265 are not applicable to the winch types specified in 1.2.

3.8 Torque clutch. Each drum provided with controlled spooling shall be equipped with a torque (slip) clutch. This clutch shall be the driving member between the drum shaft and the drum. The tension (slip) value specified below for the various winch types are based on the wire being on the average layer of the drum. The clutch shall be designed so as not to use any external energy source and shall be self compensating for normal wear.

3.8.1 The torque clutch for Type 19.5-HL-1D, 19.5-SW-1D, 15.5-SW-1D, and 15.5-SW-2D winches shall slip at a maximum of 40,000 pounds and reset at a minimum of 30,000 pounds.

3.8.2 The torque clutch for Type 8.0-SW-1D and 8.0-SW-2D winches shall slip at a maximum 20,000 pounds and reset at a minimum 15,000 pounds.

3.9 Size and weight. The winch shall be as compact and light as is consistent with adequate strength and rigidity. Weight and size shall not exceed the following:

(a) Type 19.5-HL-1D and 19.5-SW-1D winches
  (1) Maximum weight - 17,500 pounds
  (2) Maximum width (across drums) with shaft caps installed - 86 inches.
  (3) Maximum length of winch - 114 inches.
  (4) Maximum height - 85 inches
(b) Type 15.5-SW-1D Winch
  (1) Maximum weight - 16,000 pounds
  (2) Maximum width (across drums) with shaft caps installed - 86 inches
  (3) Maximum length of winch - 104 inches
  (4) Maximum height - 85 inches

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(c) Type 15.5-SW-2D Winch
(1) Maximum weight - 18,000 pounds
(2) Maximum width (across drums) with shaft caps installed - 104 inches
(3) Maximum length of winch - 104 inches
(4) Maximum height - 85 inches
(d) Type 8.0-SW-1D Winch
(1) Maximum weight - 12,500 pounds
(2) Maximum width (across drums) with shaft caps installed - 72 inches
(3) Maximum length of winch - 98 inches
(4) Maximum height - 85 inches
(e) Type 8.0-SW-2D Winch
(1) Maximum weight - 16,000 pounds
(2) Maximum width (across drums) with shaft caps installed - 104 inches
(3) Maximum length of winch - 98 inches
(4) Maximum height - 85 inches

3.10 Gearing. Gear drives shall have a service factor of 1.25.

3.11 Sheaves. All sheaves shall be designed with a minimum sheave tread diameter to wire rope diameter ratio of 30 to 1.

3.12 Special requirements.

3.12.1 Winch type 19.5-SW-1D shall be designed for one inch wire rope so as to make it identical to winch Type 19.5-HL-1D except for the controlled spooling system. The controlled spooling system on winch type 19.5-SW-1D shall be equipped to handle 7/8-inch wire.

4. QUALITY ASSURANCE PROVISIONS

4.1 In addition to the tests specified in MIL-W-17265 the tests specified in 4.2 through 4.3 shall be conducted.

4.2 Torque clutch test. Torque clutch tests shall be made at the manufacturer's plant.

4.2.1 Each drum torque clutch shall be tested by pulling a 20 foot length of wire off the drum at a speed of 240 feet per minute. The maximum tension measured during this test shall be recorded. (The use of a special locking device on the winch to hold the drum shaft stationary during this test is acceptable.)

4.2.2 The torque clutch of one winch of each type in the lot shall be tested by repeating the above test two times, waiting approximately 15 seconds between runs, and seven additional times, allowing the clutch to cool as necessary between runs. The maximum tension measured during each run shall be recorded.

4.3 Controlled spooling system and wire rope life test. One winch of each class offered for delivery on one contract or order shall be tested to show conformance with all requirements specified for the controlled spooling system (see 3.3 through 3.3.4).

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-W-17265.

6. NOTES

6.1 Ordering data. Ordering data shall be as specified in section 6 of MIL-W-17265.

6.2 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT

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